Minor Project – III (MCA-269): Literature Survey

Project	AI Tutor: An Evidence-Based Adaptive Learning System with Deliberate Scaffolding Withdrawal
Programme	Master of Computer Applications, Semester III, Batch 2025–2027
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1. Purpose and Method

This survey looks at two sets of work and compares them.

The first is research on how people learn. Over roughly forty years, repeated experiments have shown the conditions under which learning lasts. These results are well accepted in the field.

The second is the set of products and research systems used to teach technical subjects. For each one we checked which of the research findings it actually uses.

When we put the two side by side, we found that the methods known to produce lasting learning are mostly missing from the products sold to teach. The reason seems to be commercial, not technical. This gap is the main justification for our project.

We chose sources on three grounds: well-cited foundational papers, recent experiments on generative AI in learning, and commercial products available in 2026.

2. Foundational Research on Learning

2.1 Bloom's two-sigma problem

Bloom, B. S. (1984). The 2 Sigma Problem: The Search for Methods of Group Instruction as Effective as One-to-One Tutoring. *Educational Researcher*, 13(6), 4–16.

Bloom found that students who got one-to-one tutoring together with mastery learning scored about two standard deviations above students taught in a normal classroom. That means the average tutored student did better than about 98% of the classroom group. Bloom presented this as a problem: one-to-one tutoring works, but it is too expensive to give everyone, so the challenge is to find methods that get close to it for large groups.

How it relates to our project. Bloom's result sets the target. Every system in this survey, ours included, tries to get close to one-to-one tutoring at scale. We take two ideas directly from Bloom's study: mastery learning, where a student moves on only after showing mastery of the current topic, and individual pacing. Our learner model is a way of making the mastery check automatic.

2.2 The testing effect

Roediger, H. L., & Karpicke, J. D. (2006). The Power of Testing Memory: Basic Research and Implications for Educational Practice. *Perspectives on Psychological Science*, 1(3), 181–210.

Roediger and Karpicke showed that trying to recall information builds stronger memory than reading it again. Students who read a passage once and then practised recalling it remembered 61% after a week. Students who read it four times remembered 40%. The re-reading group also expected to do better, so their confidence was the opposite of their actual result.

How it relates to our project. This is why our system updates mastery only from evidence of producing something, never from reading or watching. It also explains the problem we started with: watching an instructor is re-exposure, and re-exposure builds confidence without building skill. The exercise ladder in synopsis §3.5 is a structured form of recall practice.

2.3 Desirable difficulties

Bjork, R. A., & Bjork, E. L. (2020). Desirable Difficulties in Theory and Practice. *Journal of Applied Research in Memory and Cognition*, 9(4), 475–479. **Bjork, E. L., & Bjork, R. A. (2011).** Making Things Hard on Yourself, But in a Good Way: Creating Desirable Difficulties to Enhance Learning. In *Psychology and the Real World*, 56–64.

The Bjorks separate two things: how well a student performs while practising, and how well they remember and apply it later. These often move in opposite directions. Spacing, mixing topics, changing practice conditions and testing all slow down practice and cause more mistakes, but they lead to better long-term learning. Practice that feels smooth tends to do the reverse.

How it relates to our project. Our main design rule comes from this. If some difficulty during practice helps learning, then a tutor that removes all difficulty is working against its own goal. The hint ladder, the refusal to give solution code and the planned reduction of support are all ways of keeping a useful level of difficulty. This work also explains why products avoid these methods: they feel worse in the moment, so products judged on user satisfaction are pushed away from them.

2.4 The spacing effect

Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed Practice in Verbal Recall Tasks: A Review and Quantitative Synthesis. *Psychological Bulletin*, 132(3), 354–380.

This meta-analysis covers 839 assessments from 317 experiments in 184 articles. It shows that spreading practice over time gives better retention than doing it all at once. More useful for us, it shows that the best gap between study sessions grows with how long the material needs to be remembered. So spacing is not one fixed schedule; it depends on the goal.

How it relates to our project. Our decay model is based on this. The system stores a decay rate for each skill and reduces the mastery score exponentially with the time since the last evidence. When the score drops below a threshold, the skill goes back into the revision queue. Cepeda et al. support both having a decay term and letting its rate differ by type of material.

2.5 Worked examples and the expertise reversal effect

Sweller, J., & Cooper, G. A. (1985). The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra. *Cognition and Instruction*, 2(1), 59–89. **Sweller, J. (1988).** Cognitive Load During Problem Solving: Effects on Learning. *Cognitive Science*, 12(2), 257–285.

Sweller and Cooper found that beginners who studied worked examples did better than beginners who solved the same problems alone. Solving without guidance uses up working memory on searching for a method, which leaves less capacity for actually learning the pattern. Later studies found the expertise reversal effect: as students get better, worked examples help less and less, and eventually solving problems alone becomes the better option.

How it relates to our project. Put together, these results say that support should change over time. Support that suits a beginner can hold back an intermediate student. This is why we have a five-level ladder that starts with examples and prediction tasks and ends with building from scratch, and why support is reduced on a schedule. Keeping support at one fixed level would be wrong for most of a student's progress.

2.6 Knowledge tracing

Corbett, A. T., & Anderson, J. R. (1994). Knowledge Tracing: Modeling the Acquisition of Procedural Knowledge. *User Modeling and User-Adapted Interaction*, 4(4), 253–278.

Bayesian Knowledge Tracing (BKT) treats each skill as a hidden yes/no variable (mastered or not) and updates it from the student's answers using a hidden Markov model. It has four parameters: prior knowledge, learning rate, slip probability and guess probability. It is used in the Cognitive Tutor systems and is still the standard method in this area.

How it relates to our project, and where we differ. We follow the general idea of BKT: estimate hidden skill from noisy evidence, taking context into account. We differ from classic BKT in three ways:

1. **Two scores per skill.** BKT has one mastery variable per skill. We track reading and writing ability separately because they can be very different. A profile of 10/11 on reading and 0/4 on writing averages to about 0.65, which does not tell you what the student should do next.
2. **Forgetting over time.** Classic BKT has no forgetting, so once a skill is mastered it stays mastered in the model. We add exponential decay, as described in §2.4.
3. **Only execution counts as evidence.** We use only code that was run against tests, correct predictions of program behaviour, and solved exercises. Self-reports and content completion are not used at all.

2.7 How well tutoring systems work

VanLehn, K. (2011). The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems. *Educational Psychologist*, 46(4), 197–221.

VanLehn reviewed controlled comparisons and reported two results. The effect of human tutoring (d = 0.79) is much smaller than Bloom's two sigma. Intelligent tutoring systems (d = 0.76) come very

close to human tutors. He found that the main difference between systems is the level at which they interact: systems that respond to each step or part of a step do better than those that only check the final answer.

How it relates to our project. This shapes our design. A system that only marks the final submission right or wrong loses most of the benefit. Our hint ladder responds to the student's current attempt, we detect when a student is stuck during an exercise instead of after it, and the difficulty level can change mid-task. VanLehn's numbers also show that a well-made automated tutor can come close to a human one, which makes our goal realistic.

3. Recent Studies on Generative AI in Learning

3.1 Unrestricted AI help harms learning

Bastani, H., Bastani, O., Sungu, A., Ge, H., Kabakcı, Ö., & Mariman, R. (2025). Generative AI Without Guardrails Can Harm Learning: Evidence from High School Mathematics. *Proceedings of the National Academy of Sciences (PNAS).*

This is the study closest to our project, so we describe it in some detail.

It was a randomised controlled trial with nearly 1,000 students in grades 9 to 11 at a large high school in Turkey, during the Fall 2023–24 semester. There were three groups:

Group	What they got
GPT Base	Unrestricted access to a GPT-4 chat interface while practising
GPT Tutor	GPT-4 limited to teacher-designed hints instead of answers
Control	No AI help

What they found:

- While practising with AI available, both AI groups did better than the control group.
- When AI was taken away for the exam, the students who had used unrestricted GPT-4 scored 17% lower than students who never had AI at all.
- Students used the unrestricted model as a crutch, getting answers instead of working them out.
- The GPT Tutor group, limited to hints, largely avoided this harm. They kept the benefit during practice without becoming dependent.

How it relates to our project. This study supports our main idea with a controlled experiment. It shows that unrestricted AI help damages learning while making students look and feel more successful during practice, which is the problem described in synopsis §2.2. It also shows that hint-based help, which is what we propose, reduces that harm.

We go further than the study in three ways. Our hints are spread over five levels instead of one hint policy. The limit is enforced by the server (stored hint level and output filter), not only by the prompt, since prompt-only limits can be worked around. And help is reduced over time instead of staying the same.

3.2 Completion rates in online courses

Reich, J., & Ruipérez-Valiente, J. A. (2019). The MOOC Pivot. *Science*, 363(6423), 130–131.

The authors studied 565 MOOCs from MIT and Harvard on edX over 6.5 years, with 12.67 million registrations. Completion among all registered students was 3.13% in 2017–18, down from about 4% in the two years before and almost 6% in 2014–15. 52% of registered students never started the course. Completion did not improve over six years, even among students who said they intended to finish or who paid for a verified certificate.

How it relates to our project. Online technical courses struggle even on the simple measure of completion. More important for us, none of these providers report what students who finished can actually do. Completion is the final number the industry reports, and it only measures consumption. Our system measures what students can produce and ignores completion entirely.

4. Existing Systems

4.1 Video course platforms (Coursera, Udemy, YouTube)

Aspect	What we found
Curriculum	Fixed order, same for everyone
Learner model	None
Progress measure	Percentage of content watched
Assessment	Optional quizzes, usually multiple choice, not needed to move on
Adaptivity	None
Revision support	None

Research findings used: none of those in Section 2.

These platforms are built to deliver content, and the student mostly receives it. All the problems in synopsis §2.2 apply, and the completion figures in §3.2 show the limits of this approach at scale.

4.2 Interactive coding platforms (Codecademy, freeCodeCamp, Scrimba)

Aspect	What we found
Curriculum	Fixed order, same for everyone
Learner model	None; progress is position in the course
Progress measure	Exercises completed
Assessment	Code runs against fixed tests, which is a real improvement
Adaptivity	None; same difficulty for all
Revision support	None

Research findings used: recall practice (§2.2), partly, since students write code instead of watching.

This is a clear step up from video and the closest category to our system. Three issues remain. Support is heavy and never reduced: exercises usually give a partly written file with instructions at every step, which is level 3 of our ladder kept for the whole course, and by §2.5 that is the wrong level for anyone past the beginner stage. There is no model of the student, so someone who already knows a topic still has to go through it. Earlier topics are never revisited.

4.3 Practice and assessment platforms (LeetCode, HackerRank, Exercism)

Aspect	What we found
Curriculum	None; a searchable problem list
Learner model	Number of problems solved and success rate; no skill-level estimate
Progress measure	Problems solved, contest rating
Assessment	Code runs against thorough test suites, which is strong
Adaptivity	Difficulty tags set by editors, not by the student's level
Revision support	None

Research findings used: recall practice (§2.2), at full difficulty.

These sites have the strongest assessment in the survey but do no teaching. They assume the student already knows what to practise and can already build solutions alone, so they really serve students at level 5 of our ladder. For a student at level 1 or 2 they are the empty-file problem in its most direct form. This matters for DSA in particular, since these are the sites most students use for it. There is no prerequisite structure and no sense of whether a student is ready for a problem.

4.4 Spaced-repetition tools (Anki, Quizlet)

Aspect	What we found
Curriculum	Card decks made by users
Learner model	Ease factor and interval for each card, which is a real forgetting model
Progress measure	Cards due, retention rate
Assessment	Student reports how hard recall was
Adaptivity	Review interval adjusts per card
Revision support	Strong; this is what the product is for

Research findings used: spacing (§2.4) and recall practice (§2.2), both done well.

This is the only category that handles forgetting and revision properly. Two limits stop it from solving our problem. Assessment is self-reported: the student decides whether they recalled correctly, which §2.2 shows is unreliable. Cards test facts, not building ability, so no amount of Anki practice shows whether someone can write a working program. Cards are also made by hand.

We take Anki's scheduling idea, replace self-reporting with evidence from running code, and replace fact cards with building tasks.

4.5 General AI assistants (ChatGPT, Claude, GitHub Copilot)

Aspect	What we found
Curriculum	None
Learner model	Nothing kept between sessions
Progress measure	None
Assessment	None
Adaptivity	Responds only to the current request
Revision support	None

Research findings used: none, and several are directly contradicted.

The Bastani et al. trial (§3.1) tested this kind of tool directly and found a 17% drop in performance once access was removed, compared with no AI help at all. The reason is clear: an assistant that answers on request removes the need to recall (§2.2) and removes useful difficulty (§2.3), while giving a strong feeling of progress. These products are also judged on engagement and helpfulness ratings, which reward giving answers.

4.6 AI tutoring products (Khanmigo, Duolingo Max)

Khanmigo from Khan Academy is the closest commercial product to ours. It uses Socratic questioning and is designed to guide instead of giving answers. As of March 2026 it is available to free users and used by teachers in more than 44 countries.

Duolingo Max adds AI explanations ("Explain My Answer") and role-play conversations to Duolingo's gamified course, for $29.99 per month.

Aspect	Khanmigo	Duolingo Max
Curriculum	Khan Academy's existing linear course	Fixed gamified sequence
Learner model	Progress per exercise within the platform	Streaks, XP, position in the unit
Refuses to give answers	Yes, through the prompt	No
Support reduced over time	No	No
Assessment	Existing exercise checking	Answer matching
Revision support	Limited	Streaks, which drive engagement, not spaced revision

Research findings used: Khanmigo partly uses the Socratic approach (§2.3). Neither uses §2.4, §2.5 or evidence-only assessment.

Khanmigo shows that a tutor which does not give answers can work commercially at large scale, which lowers the risk of our central idea. There are still three differences. As far as we can tell its limit is set in the prompt, which a determined user can get around; ours is enforced on the server and by an output filter. Its support does not reduce over time, which §2.5 says is wrong for a student who is improving. And it sits on top of a linear course instead of a prerequisite graph, so it cannot skip known topics or choose the next step from the student's level.

We included Duolingo Max because it shows the incentive problem most clearly. Streaks and XP are there to bring users back every day, and its known limits (short scripted conversations, no pronunciation scoring) suggest a product built for daily return more than for final ability.

5. Comparison Summary

Which research findings each category uses:

Category	**Recall practice (§2.2)**	**Desirable difficulty (§2.3)**	**Spacing (§2.4)**	**Fading support (§2.5)**	**Learner model (§2.6)**	**Step-level help (§2.7)**	**Evidence-based assessment**
Video courses	No	No	No	No	No	No	No
Interactive coding	Partly	No	No	No	No	Partly	Yes
Practice platforms	Yes	Yes	No	No	No	No	Yes
Spaced repetition	Yes	Yes	Yes	No	Partly	No	No
General AI assistants	No	No	No	No	No	Partly	No
AI tutoring products	Partly	Partly	No	No	Partly	Partly	Partly
Our system (planned)	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

6. Research Gap

We draw three conclusions from the survey.

Each method we use already has research behind it. Recall practice, spacing, desirable difficulty, fading support, knowledge tracing and step-level interaction are not new. Each has decades of

experimental support. Our contribution is to combine all of them in one system, which the table above shows has not been done.

Two of our design choices are new, and both are based on evidence. Keeping reading and writing mastery separate does not appear in the knowledge tracing literature, which uses one score per skill. We do it because the two scores can differ so much that an average gives a misleading next step. Reducing support on a planned schedule, as part of the student's plan and not only on request, follows from the expertise reversal effect, but none of the systems we surveyed does it.

The gap is mainly commercial. None of the methods in Section 2 is hard to build. Spacing is one exponential term, and a hint ladder is a number stored on the server. Products leave them out because each one lowers engagement: the product feels harder, and users finish and leave sooner. A product judged on time in the app, daily active users or satisfaction ratings would hurt its own numbers by adopting them.

This is why a new product can take this approach where existing ones cannot. We treat time spent in the app as something to reduce, as long as students gain the same ability. The Bastani et al. trial suggests that ignoring this costs students about 17% of what they would otherwise retain.

7. References

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